ABSOLUTE / UPAJ

Farm OS: IoT, GIS and GrowStac product context

IoT and GIS monitoring for indoor and open-field farms, alongside GrowStac’s current public proposition.

Role: Product leadership across the Absolute portfolio | Period: October 2021–November 2024

Reported outcomes and scope

IoT + GIS

Documented monitoring capabilities for controlled environments and open fields.

Two operating contexts

Indoor sensor/controller information and parcel-level open-field monitoring.

The product problem

Controlled-environment agriculture and open-field farming have different monitoring needs. An indoor grower works with sensors and controllers, while an open-field grower needs a view of crops and conditions at parcel level. A useful system has to respect those differences.

My role

The Professional Profile documents product work on a Farm OS within the Upaj portfolio. My contribution is described at the product level: connecting the operating requirement with the monitoring capabilities and the farmer-facing experience.

What I built

The system combines indoor IoT sensors with GIS-based crop monitoring for outdoor parcels. This brings different inputs into a broader farm-management stack while retaining the needs of each setting.

The monitoring capability sits alongside advisory, market information and engagement work. Institutional presentations provide additional product context through farm health, satellite information and field-level mapping.

Two operating environments, different inputs

I worked on Farm OS as the monitoring layer spanning indoor controlled environments and open-field farms. IoT sensors supported indoor controllers, while the GIS-based app supplied parcel-level crop monitoring outdoors. Bringing them into a broader software stack did not mean that the same data view or hardware applied to every farm.

The indoor product connected with sensor-equipped growing conditions. The open-field app used parcel-level geographic information to support near-real-time monitoring. My product work brought those information sources into the farm-management experience, with monitoring kept distinct from claims of automated physical control.

GrowStac: current product reference

Upaj's GrowStac page, captured in October 2026, describes precision agriculture through advisory, technology, research and personalised solutions. Its panels connect IoT, satellite and soil-health data with farm insights, API integration and agronomy support. This current description helps show the wider product setting for farm monitoring.

The public proposition includes enterprise, NGO and cooperative support, app-based advisory checklists and tailored research around crop yield and quality. These are 2026 product descriptions, separate from my historical Farm OS contribution. The public page does not demonstrate every live monitoring workflow or establish an individual yield gain caused by the system.

Monitoring as part of a usable farm service

The historical Farm OS work sits beside the Upaj engagement revamp and crop-health advisory. Monitoring supplied information about the farm; the farmer experience still needed to make that information usable. The portfolio therefore connects sensing and maps with the broader interface and advisory work, while keeping product-specific results distinct.

What changed

The delivered scope described in the portfolio is an indoor/open-field monitoring system using IoT and GIS inputs. Its role was part of the wider farm service: give the user a view of farm conditions that could sit alongside advisory and other useful information. No separate Farm OS adoption or productivity figure is disclosed.

The lesson from this work

This case demonstrates product design around operating environments. The common problem is visibility, but the inputs, decision cycle and controls differ between an indoor farm and an open field.

Watch the current public product reference

These recordings capture Upaj's current public product information in October 2026. They provide visual context for the historical contributions described here; they do not establish ownership of every current feature.

Visual references and walkthroughs

Upaj: Growstac advisory

Upaj: Growstac advisory Public reference captured on 13 October 2026.

Upaj: Growstac demo inquiry

Upaj: Growstac demo inquiry Public reference captured on 13 October 2026.

Upaj: Growstac hero

Upaj: Growstac hero Public reference captured on 13 October 2026.

Upaj: Growstac inquiry interests

Upaj: Growstac inquiry interests Public reference captured on 13 October 2026.

Upaj: Growstac personalised

Upaj: Growstac personalised Public reference captured on 13 October 2026.

Upaj: Growstac research

Upaj: Growstac research Public reference captured on 13 October 2026.

Upaj: Growstac technology

Upaj: Growstac technology Public reference captured on 13 October 2026.

Upaj GrowStac

Generated GrowStac overview from captured public pages and original website illustrations. Public product reference captured on 13 October 2026.

Documented workflow: Indoor sensors / outdoor parcels → IoT readings and GIS signals → Farm monitoring view → Conditions and crop context → Operational next actions

An authored explanation of the workflow described in this case study. This is a diagram, not a captured product screen.

View the online case study for the full image gallery and playable videos.

Evidence and attribution

Institutional decks and current public product captures are supporting references; their scope is not a standalone performance result.

GrowStac reference is a 13 October 2026 public product capture after the documented Absolute tenure. Monitoring, recommendation and automated control are distinct; no isolated Farm OS adoption, yield or cost result is supplied.